SMART WASTE MANAGEMENT SYSTEM

Python Program

```
import json
import csv
from datetime import datetime
```

(a) Base class and derived classes

```
class WasteClient:
def __init__(self, client_id, name):
self.client_id = client_id
self.name = nam
def calculate_bill(self, weight, rate):
return weight * rate
class Household(WasteClient):
waste_type = "Household"
class Industry(WasteClient):
waste_type = "Industry"
class Hospital(WasteClient):
waste_type = "Hospital"
def check_compliance(self):
return "Biomedical waste compliance checked"
```

(b) Schedule Manager using JSON

```
class ScheduleManager:
def __init__(self, filename):
self.filename = filename
def load_schedule(self):
with open(self.filename, "r") as file:
return json.load(file)
def display_schedule(self):
schedules = self.load_schedule()
print("\n--- COLLECTION SCHEDULE ---")
for s in schedules:
print("Client ID :", s["client_id"])
print("Name      :", s["name"])
print("Type      :", s["type"])
print("Date      :", s["date"])
print("Time      :", s["time"])
print()
```

(c) Load rate table from CSV

```
def load_rates(filename):
rates = {}
with open(filename, "r") as file:
reader = csv.DictReader(file)
for row in reader:
rates[row["type"]] = float(row["rate_per_kg"])
return rates
```

(d) Generate receipts in text files

```
def generate_receipt(client, weight, rate):
```

```
amount = client.calculate_bill(weight, rate)
filename = client.client_id + "_receipt.txt"
with open(filename, "w") as file:
file.write("SMART WASTE MANAGEMENT SYSTEM\n")
file.write("--------------------------------\n")
file.write("Client ID : " + client.client_id + "\n")
file.write("Name      : " + client.name + "\n")
file.write("Type      : " + client.waste_type + "\n")
file.write("Weight    : " + str(weight) + " kg\n")
file.write("Rate      : Rs." + str(rate) + " per kg\n")
file.write("Amount    : Rs." + str(amount) + "\n")
if isinstance(client, Hospital):
file.write("Compliance: " +
client.check_compliance() + "\n")
file.write("--------------------------------\n")
print("Receipt generated:", filename)
return amount
```

(e) Log collection events

```
def log_event(client, weight, amount):
with open("collection_log.txt", "a") as file:
time = datetime.now().strftime("%Y-%m-%d %H:%M:%S")
file.write(
time + " | " +
client.client_id + " | " +
client.name + " | " +
client.waste_type + " | " +
str(weight) + " kg | Rs." +
str(amount) + "\n"
)
```

(f) Generate summary report

```
def generate_report(records):
total_weight = 0
total_amount = 0
with open("summary_report.txt", "w") as file:
file.write("SMART WASTE MANAGEMENT - SUMMARY REPORT\n")
file.write("----------------------------------------\n")
for record in records:
file.write(
record["client_id"] + " | " +
record["name"] + " | " +
record["type"] + " | " +
str(record["weight"]) + " kg | Rs." +
str(record["amount"]) + "\n"
)
total_weight += record["weight"]
total_amount += record["amount"]
file.write("----------------------------------------\n")
file.write("Total Waste   : " +
str(total_weight) + " kg\n")
file.write("Total Billing : Rs." +
```

```
str(total_amount) + "\n")
print("\n--- SUMMARY REPORT ---")
print("Total Waste   :", total_weight, "kg")
print("Total Billing : Rs.", total_amount)
```

Main program

```
schedule_manager = ScheduleManager("schedule.json")
schedule_manager.display_schedule()
rates = load_rates("waste_rates.csv")
schedules = schedule_manager.load_schedule()
records = []
for s in schedules:
client_id = s["client_id"]
name = s["name"]
waste_type = s["type"]
if waste_type == "Household":
client = Household(client_id, name)
weight = 10
elif waste_type == "Industry":
client = Industry(client_id, name)
weight = 25
elif waste_type == "Hospital":
client = Hospital(client_id, name)
weight = 15
else:
continue
rate = rates[waste_type]
amount = generate_receipt(client, weight, rate)    log_event(client, weight, amount)
records.append({
"client_id": client.client_id,
"name": client.name,
"type": client.waste_type,
"weight": weight,
"amount": amount
})
generate_report(records)
print("\nAll operations completed successfully.")
```

Sample waste_rates.csv

```
type,rate_per_kg
Household,10
Industry,20
Hospital,50
```

Sample schedule.json

```
[
{
"client_id": "H001",
"name": "Arun Home",
"type": "Household",
"date": "25-08-2026",
"time": "09:00"
```

```
},
{
"client_id": "I001",
"name": "ABC Factory",
"type": "Industry",
"date": "25-08-2026",
"time": "11:00"
},
{
"client_id": "P001",
"name": "City Hospital",
"type": "Hospital",
"date": "25-08-2026",
"time": "14:00"
}
]
```


Output

```
Client ID : H001
Name      : Arun Home
Type      : Household
Date      : 25-08-2026
Time      : 09:00
Client ID : I001
Name      : ABC Factory
Type      : Industry
Date      : 25-08-2026
Time      : 11:00
Client ID : P001
Name      : City Hospital
Type      : Hospital
Date      : 25-08-2026
Time      : 14:00
Receipt generated: H001_receipt.txt
Receipt generated: I001_receipt.txt
Receipt generated: P001_receipt.txt
Total Waste   : 50 kg
Total Billing : Rs. 1350.0
All operations completed successfully.
```